HIT: A fully automated mosaicking software for corneal confocal microscopy

Supplementary File

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1 HIT Startup Options

Starting *HIT* in from a file explorer GUI opens the *HIT* main dialog GUI.

On the command line, there are the following options. Here, `<executable>` is a placeholder for the executable file name (normally, this is `HITgui.exe` on Windows systems or `HITgui` on Linux systems):

- "`<executable>`" opens the *HIT* main dialog GUI.
- "`<executable> -p`" opens the *HIT* parameter dialog GUI.
- "`<executable> -a <filepath>`" starts *HIT* in automation mode; this opens the main dialog GUI, automatically processes the dataset specified by `<filepath>` and closes the GUI after finishing the dataset.

2 HIT Main Dialog GUI

The *HIT* main dialog GUI (see Figure 1) controls the HIT algorithm workflow.

The button “Add” lets the user choose one or more dataset files to add to the list of datasets. Datasets can be removed from the list by selecting them in the list (multi-select using “Shift”+”Click” is supported) and clicking the button “Remove”.

The button “Start” starts the workflow, which processes all datasets in the current datasets list. This locks all GUI elements for the user except for the “Start” button itself; its label is changed to “Stop”. Pressing the button again during processing interrupts the execution; it can take some time before the process actually stops and the GUI controls become unlocked.

The process log provides a log on the current progress.

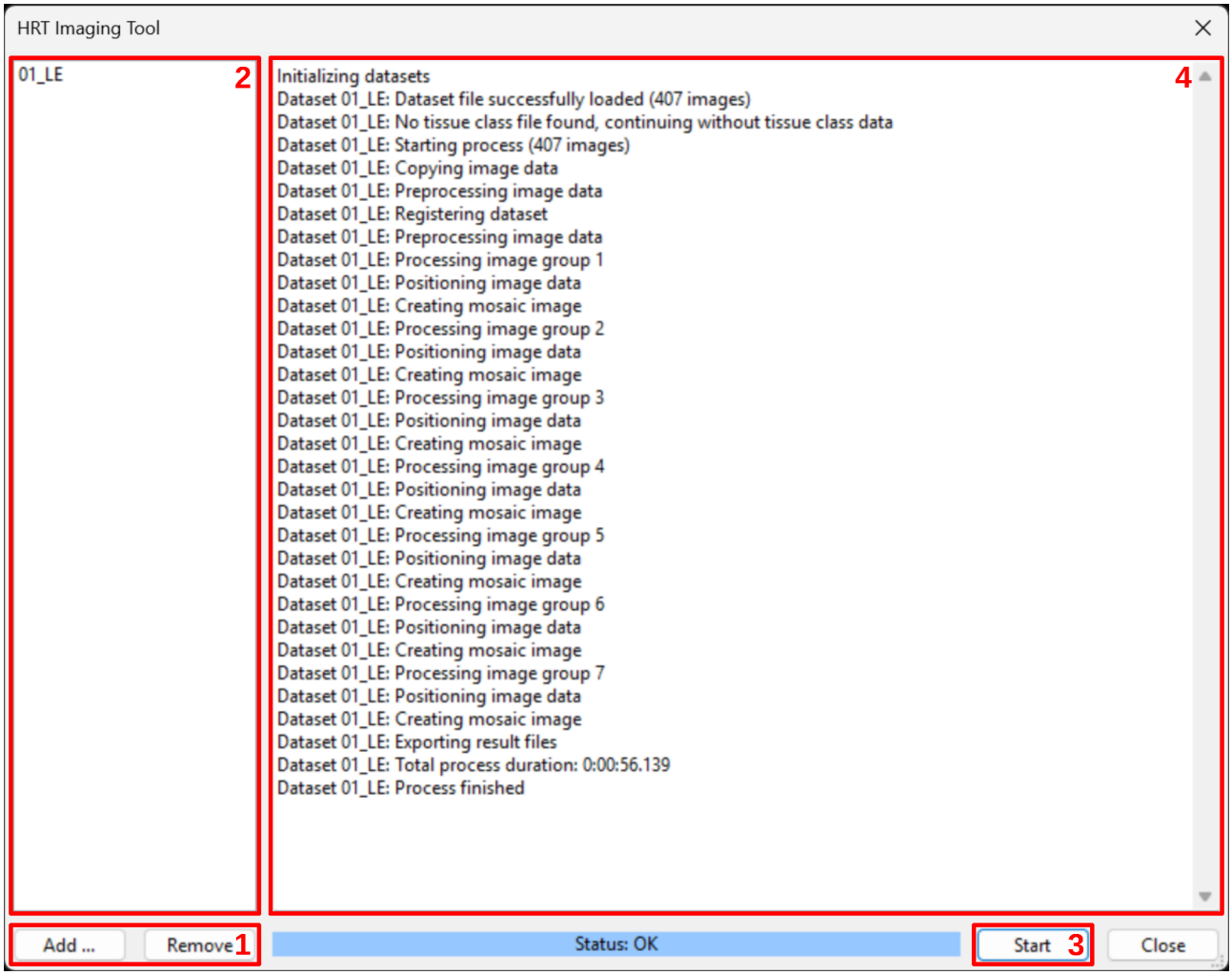

Figure 1: *HIT* main dialog GUI. The main GUI elements are highlighted in red. (1) Buttons to add datasets to or remove them from the dataset list. (2) List of datasets to be processed. (3) Button to start the mosaicking process. (4) Process log.

3 HIT Parameter Dialog GUI

The *HIT* parameter dialog GUI (see Figure 2 on the next page) provides access to all *HIT* process parameters.

The button “Defaults for 2D datasets” reverts all parameters to their default values.

The process parameter area provides appropriate controls to manipulate all *HIT* process parameters. The most relevant process parameters are detailed in the Section 4.

> **Note:** Clicking the name or the control of any parameter brings up a short description of the respective parameter in the text area below the parameter list.

The buttons “OK” and “Cancel” both close the dialog GUI, either saving or discarding all applied changes. A warning message appears if changes are about to be discarded.

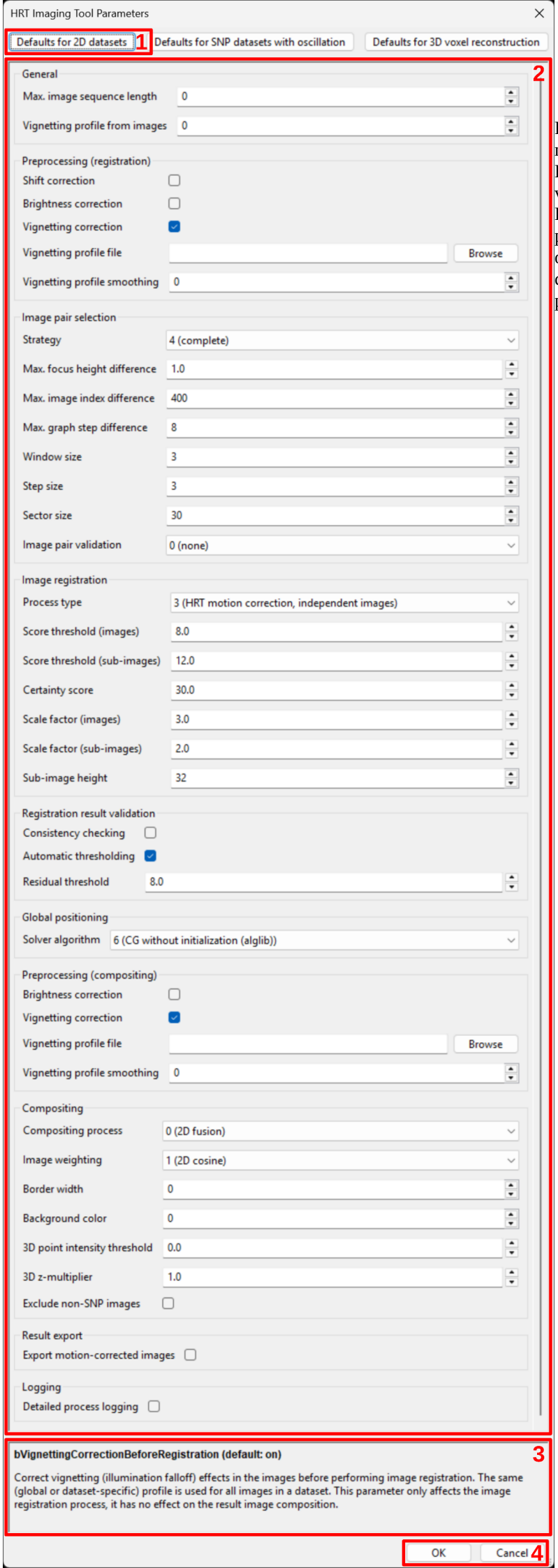

Figure 2: *HIT* parameter dialog GUI. The main GUI elements are highlighted in red. (1) Button to revert all parameters to their default values. (2) Process parameter area. (3) Description of the currently focused parameter (the focus is changed by clicking on a parameter name or its respective control). (4) Buttons to save or discard the parameter changes.

4 Process Parameters

The following list explains the most relevant *HIT* process parameters and refers to the *HIT* source code classes that implement the respective features:

Vignetting correction: This switch controls the inclusion of a vignetting correction preprocessing step in the pipeline. Vignetting correction can be controlled separately for the image registration and for the mosaic montage. If used, a vignetting correction profile can either be specified explicitly from an image file or calculated from each dataset separately, using the average image from the entire dataset. This is implemented in the `CVignettingCorrection` class.

Vignetting profile file: This parameter is only considered if the *Vignetting correction* switch is turned on. It specifies the file path of a vignetting profile image that is used to correct the illumination fall-off effect. If left empty, the vignetting profile is instead calculated as the average image from the entire dataset. Vignetting correction is implemented in the `CVignettingCorrection` class.

Scale factor: These values (separate for the image level and the sub-image level registration) specify the respective factors for down-scaling the image sizes used for the computationally expensive image registration process. A value of 1 means no scaling is performed; values larger than 1 activate downscaling by the respective factor. This is implemented in the `CRegStepImageScalation` class.

Image pair selection strategy: This parameter controls which image pairs are processed during image registration. The default value "*complete*" describes an exhaustive image pairing, using all possible image pairs (i, j), $i<j$ for registration. The opposite end of the complexity spectrum is marked by the "*simple forward*" strategy, which only registers image pairs $(i, i+1)$. The image pair selection strategy is implemented in the `CRegistrationStrategy` class and its derived classes.

Image registration process type: This parameter specifies whether the image data represent a continuous sequence (recorded continuously with 30 images per second) or a non-continuous set of independent images. A third option specifies to use rigid image registration only, without motion correction. This parameter predominantly controls the setup of the image registration pipeline in the `CImageRegistrator` class and the specific regularization technique used in the motion trajectory calculation step, which is implemented in the `CHRTGlobalPositioning` class.

Sub-image height: This value specifies the height (i.e. the number of pixel rows) of the sub-images used for sub-image decomposition and registration. It is not used if the *Image registration process type* is specified as rigid. The sub-image size directly affects the potential to resolve motion changes during the motion estimation and correction tasks, as each sub-image defines a support point on the estimated motion trajectory. Therefore, smaller sub-image size increases the resolution for motion estimation, but smaller sub-images can also reduce the robustness of the sub-image registration task. The default value of 32 has been established as a good compromise and should only be changed with care. The height of each single raw image of the dataset must be a multiple of the sub-image height. The sub-image registration is implemented in the `CSubImageCorrelator` class.

Score threshold: These threshold values (separate for the image level and the sub-image level registration) specify the basic result validation scheme for the image (or sub-image) registration results. Results with a score value above the threshold are kept, below-threshold result are discarded immediately. Registration result validation is implemented in the `CCorrelationOffsetClassifier` class.

Automatic thresholding: This switch turns the automatic score threshold calculation on or off. If switched off, the static score threshold values above are used for registration result validation. It is advised to turn automatic thresholding on by default, and only resort to static threshold settings in cases where the automatic method produces unsatisfactory results. The automated threshold calculation is implemented in the `CRegStepScoreThresholdAdapter` and `CRegStepSubImageScoreThresholdAdapter` classes and their derived classes.

Residual threshold: This threshold value controls the automatic score threshold calculation. It is only used if the *Automatic thresholding* switch is turned on. This parameter specifies the highest acceptable residual error for any single image pair or sub-image registration result. The residual error for a single registration result is defined as follows: if $\boldsymbol{v}$ is the registration result vector for any given image (or sub-image) pair and $\boldsymbol{\Delta p}$ is the position difference vector of the same image (or sub-image) pair after the motion trajectory calculation step, then the length of the residual vector $\boldsymbol{e} = \boldsymbol{\Delta p} - \boldsymbol{v}$ is the residual error for the respective single registration result. During automatic score threshold calculation, the score threshold value is increased continuously, thereby discarding all image (or sub-image) registration results with a smaller registration score value. This process stops as soon as the largest remaining registration result residual error is below the residual threshold parameter. Increasing this parameter increases the chance for larger connected mosaic images, but also the risk of deformed mosaic images (due to remaining false registration results that are included in the motion trajectory calculation step), and vice versa. The automated threshold calculation is implemented in the `CRegStepScoreThresholdAdapter` and `CRegStepSubImageScoreThresholdAdapter` classes and their derived classes.

Export motion-corrected images: This switch controls whether the motion-corrected and globally aligned images are exported separately as image files. This setting can be used for in-depth result analysis if required, and it should be turned on for volume reconstruction use cases. The file export is implemented in the `CSNPDatasetResultLoader` class.

Exclude non-SNP images: This switch controls the inclusion of tissue class information in the mosaic image montage process. This parameter requires the presence of tissue class information (see Section 2.2.3 in the main paper). This parameter is used in the `CCompositingFusion` class.

5 Runtime Performance

The runtime performance of *HIT* heavily depends on the process parameters. In most real-world use cases (and for most parameter settings), the pairwise image registration step is the most time-consuming algorithmic processing step in the pipeline. With the recommended default parameters, all possible image pairs are considered for this registration step, resulting in a more or less quadratic runtime performance regarding the dataset size (in terms of the number of images). Figure 3 provides a scatter plot of the total dataset processing times for 106 datasets (source: Badian et al. 2021. doi:10.6084/m9.figshare.14481249), measured on a common Windows PC (Intel Core i9-11900K CPU). The approximately quadratic total dataset runtime behavior (with respect to dataset size) is in line with the quadratic number of registered image pairs with the default exhaustive image pair selection strategy (see Section 4).

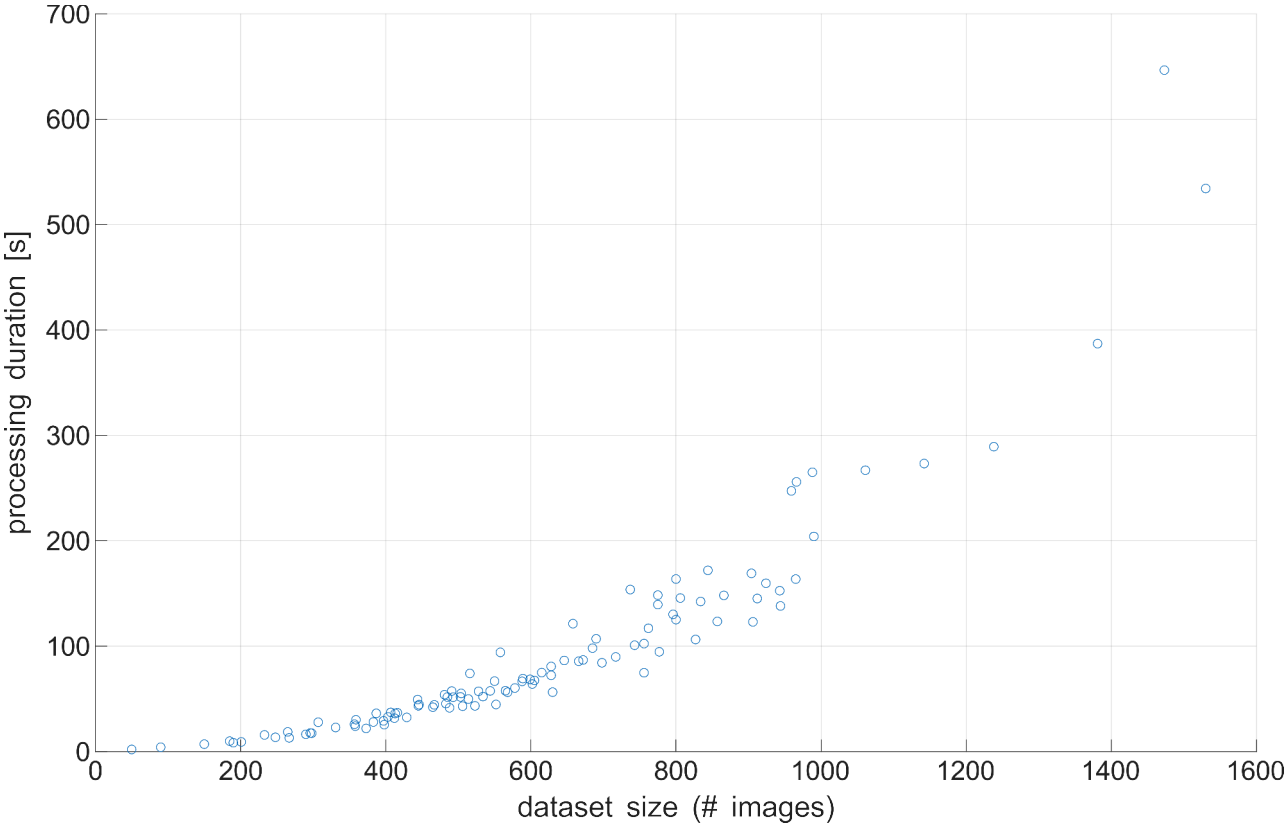

Figure 3: *HIT* runtime performance plot. Each data point is the total dataset processing time (in seconds) on a common Windows PC (Intel Core i9-11900K CPU) using the default *HIT* process parameters. The datasets are publicly available (source: Badian et al. 2021. doi:10.6084/m9.figshare.14481249).